

Training scan x_i

scanner b_i, label y_i

Scanner model

label-protected ComBat
fit on training data:
γ_b, δ_b per connection

Oracle (upper bound)

true scan at the
other scanner



Counterfactual $\tilde{x}_i$

scanner $1 - b_i$, same y_i

Model *f*



shared weights

Training loss

$\mathcal{L} = \mathrm{CE}(f(x_i), y_i) + \mathrm{CE}(f(\tilde{x}_i), y_i)$

At test: new sites are used as they are
(no harmonisation, no counterfactuals)